$\phi : G \to H$ is an isomorphism

iff 1) ϕ is bijective

2) $\forall\, a, b \in G \quad \phi(ab) = \phi(a)\phi(b)$

$\forall\, a \in G \qquad \phi(a^{-1}) = \phi(a)^{-1}$

Theorems 9.7 $\longrightarrow$ 9.9

Details as exercises.

Key ideas.

<u>Theorem 9.7</u> If $|G| = \infty$ and G is cyclic

i.e. $G = \langle g \rangle$, for some $g \in G$.

Claim: $G \cong \mathbb{Z}$, $\mathbb{Z}$ = additive group of integers.

The isomorphism $\phi : G \to \mathbb{Z}$

is defined by $\phi(g^k) = k$

Details are confirming that ϕ is bijective and satisfies the hom. prop.

<u>Theorem 9.8</u> The similar mapping $\phi : G \to \mathbb{Z}_n$

$$\phi(g^k) = \bar{k}.$$

<u>Corollary 9.9</u> We've already proved (exercise to chap 6 Lagranges) that if

$|G| = p$, a prime. Then G is cyclic of order p. So by Theorem 9.8 $G \cong \mathbb{Z}_p$.

<u>Theorem 9.10</u> Isomorphism is an equivalence relation on groups.

<u>Proof</u> We need to show $\cong$ is reflexive, symmetric and transitive.

1. Claim: ~~∀~~ For any group G, $G \cong G$.

<u>Pf</u>: The isomorphism $\phi : G \to G$ is the <u>identity transformation</u> defined by $\forall a \in G$ $\phi(a) = a$.

This is a bijection

$\forall a, b \in G$ $\boxed{\phi(ab)} = ab$

$= \boxed{\phi(a)\phi(b)}$

So this is an isomorphism, and so $G \cong G$.

2. Claim. If $G \cong H$ then $H \cong G$.

<u>Pf</u>: already done in part (1) of Theorem 9.6. ϕ^{-1} is the required isomorphism $H \to G$.

3. Transitivity.

Claim: If ($G \cong H$ and $H \cong K$) then $G \cong K$.

Pf Let's say $\phi : G \to H$ and $\psi : H \to K$ are isomorphisms.

The required isomorphism $G \to K$ is the map $\psi \circ \phi$.

$$\begin{tikzcd} G \arrow[r, "\phi"] \arrow[rr, bend right, "\psi \circ \phi"'] & H \arrow[r, "\psi"] & K \end{tikzcd}$$

ψ: psi

ϕ = phi

1. From set theory / mapping theory

$\psi \circ \phi$ is bijective since ϕ, ψ are.

2. For any $a, b \in G$

$$\begin{aligned} (\psi \circ \phi)(ab) &= \psi(\phi(ab)), \text{ def of comp.} \\ &= \psi(\phi(a)\phi(b)), \text{ hom prop for } \phi. \\ &= \psi(\phi(a))\,\psi(\phi(b)), \text{ hom prop for } \psi \\ &= (\psi \circ \phi)(a)\,(\psi \circ \phi)(b), \text{ def of composition.} \end{aligned}$$

Therefore $\psi \circ \phi$ is an isomorphism and $G \cong K$ as required. ∎

Cayley's Theorem Theorem 9.12

Any group G is isomorphic to a group of permutations

<u>Pf.</u> Observation: In a Cayley table for G we see that <u>each element</u> of G is naturally associated to a permutation of the elements of G, namely the row of the table corresponding to <u>that element</u>.

For any $g \in G$ consider the map $\lambda_g : G \to G$ defined by $\forall a \in G \ \lambda_g(a) = ga$

So λ_g is "multiplication on the left by g".

	a
g	ga

From basic group properties can

Show that $\lambda_g : G \to G$ is bijective, or in other words λ_g is a permutation of G.

Consider the permutation group

$$\overline{G} = \{ \lambda_g : g \in G \}.$$

<u>Claim:</u> $\overline{G}$ is a group under the operation of <u>composition</u>.

1. The identity of $\overline{G}$ is the map λ_e

2. Closure

$$\lambda_a \circ \lambda_b = \lambda_{\underline{ab}}$$

$\underline{ab}x$

3. Associative.

well composition is universally associative.

4. $(\lambda_g)^{-1} = \lambda_{g^{-1}}$

These show $\overline{G}$ is a group

Then the required isomorphism will be $\phi : G \longrightarrow \overline{G}$ defined by $g \longmapsto \lambda_g$

<u>Observations</u>

1). Initially, this seems simplifying, we only need to study permutation groups.

But really: it means permutation groups are just as complicated as the space of all groups

2). This shows us that we can calculate with groups on the computer.

Rules for calculating with permutations can be easily implemented in software.

Continue at 5:07.

9.2. Direct products.

1. External direct products

From basic set theory, given any two sets G, H there is a natural way to build a new set from these

$$G \times H = \{ (g, h) : g \in G, h \in H \}$$

called the Cartesian product of G, H.

eg. the plane $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R} = \{ (x, y) : x, y \in \mathbb{R} \}$.

If F is the set of teams then $F \times F$ is the set of games in a typical double round robin season.

We can use this in group theory.

Given two groups $(G, \cdot)$, $(H, \circ)$ we can build a new group on the set $G \times H = \{ (g, h) : g \in G, h \in H \}$.

where the operation on $G \times H$ is defined in the natural way

$$(g_1, h_1) \cdot (g_2, h_2) = (g_1 \circ g_2, h_1 \circ h_2)$$

$$\in G \qquad \in H \qquad \in G \times H$$

or written multiplicatively

$$(g_1, h_1)(g_2, h_2) = (g_1 g_2, h_1 h_2)$$

<u>Prop 9.13</u> checking all the details $G \times H$ is a group under this operation.

called "the (external) direct product of G and H"

Seen this in linear algebra in "direct sums of vector spaces"

<u>Proof</u> (main points)

- Clearly $G \times H$ is closed under this operation
- the identity element is

3. Associative, which follows from the associativity in G and H

4. Given $(g, h) \in G \times H$. Its inverse will be

$$(g, h)^{-1} = (g^{-1}, h^{-1})$$

So $G \times H$ is a group.

<u>9.15</u> $\mathbb{Z}_2 = \{0, 1\}$

Consider $\mathbb{Z}_2 \times \mathbb{Z}_2 = \{(0,0), (0,1), (1,0), (1,1)\}$.

Actually can show that

$$\mathbb{Z}_2 \times \mathbb{Z}_2 \cong U(8), \text{ (exercise).}$$

Concept extends in obvious / inductive way to direct products of any number of components.

$$\prod_{i=1}^{n} G_i = G_1 \times G_2 \times \dots \times G_n$$
$$= \{(g_1, g_2, \dots, g_n) : g_i \in G_i\}$$

Theorem 9.17 For $(g, h) \in G \times H$ if $|g|, |h|$ are both finite then

$$|(g, h)| = \operatorname{lcm}(|g|, |h|).$$

Proof: Exploiting the ramifications of the rule

$$(g, h)^n = (g^n, h^n) \quad \square$$

Leads to

Theorem 9.21

$$\mathbb{Z}_m \times \mathbb{Z}_n \cong \mathbb{Z}_{mn}$$

iff $\gcd(m, n) = 1$

Proof uses theorem 9.17.

since $\gcd(m,n) = 1 \Rightarrow \operatorname{lcm}(m,n) = mn$

Internal direct product

- External direct products are a tool for building complicated from a collection of component groups.

- Internal direct products is a concept for recognising when a complicated group can be expressed as a direct product of a collection of its subgroups.

<u>Idea</u> Take a pair of subgroups H, K of G, and we'll model the ordered pairs $(h, k) \in H \times K$ using the products $hk \in HK \subset G$.

where

$$HK := \{ hk : h \in H, k \in K \}.$$

The technical conditions that will imply that $G \cong H \times K$, the external direct product.

are

1). $HK = G$

2) $H \cap K = \{e\}$

3) $hk = kh$, for every $h \in H$, $k \in K$.

$H \cap K = \{e\}$, this ~~is~~ is needed to avoid "double representation"

Suppose $h_1 k_1 = h_2 k_2$

$$\iff \underset{\substack{\cap \\ H}}{h_2^{-1} h_1} = \underset{\substack{\cap \\ K}}{k_2 k_1^{-1}}$$

$\iff x \in H \cap K$.

But $x = e \Rightarrow h_2^{-1} h_1 = e \Rightarrow \underline{h_1 = h_2}$

$k_2 k_1^{-1} = e \Rightarrow \underline{k_2 = k_1}$

$hk = kh$ is needed to make the "products" work.

$$(h_1, k_1)(h_2, k_2) := (h_1 h_2, k_1 k_2)$$

$$\underbrace{h_1 k_1 \quad h_2 k_2}_{\in G} = h_1 k_1 h_2 k_2$$

$$= \underbrace{h_1 h_2}\ \underbrace{k_1 k_2}$$

these conditions of internal direct product

give the proof of theorem 9.27.